Input system:
Dependent variables:$y_5, y_4, y_6, y_3, y_1, y_2$
Independent variables:a, x_1, x_3, b, x_2
Functions:

$$y_1 = x_3^3 \sin\left(\cos\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + x_3^3 b\ln\left(x_2\right)^4 + \sin\left(a\ln\left(x_2\right)^2 + bx_3\right)^5 + \mathrm{Dv}\left(g^2, 1, (-1)\,; x_1\right)$$

$$y_2 = \sin\left(\cos\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + b\ln\left(x_2\right)^4 + \sin\left(a\ln\left(x_2\right)^2 + bx_3\right)^5 + \mathrm{Dv}\left(g^2, 1, (-1)\,; x_1\right)$$

$$y_3 = x_2^2 \cos\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + \cos\left(a\ln\left(x_2\right)^2 + bx_3\right)^4 + \mathrm{EK}\left(\sin\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right), x_1\right)^2$$

$$y_4 = x_2^2 \cos\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right) + \cos\left(a\ln\left(x_2\right)^2 + bx_3\right)^4 + \mathrm{EK}\left(\sin\left(\sin\left(a\ln\left(x_2\right)^2 + bx_3\right)\right), x_1\right)$$

$$y_5 = x_1^2 \mathrm{ch}\left(a\ln\left(x_2\right)^2 + bx_3\right)^2 + x_1^2 \sin\left(a\ln\left(x_2\right)^2 + bx_3\right) + \mathrm{sh}\left(a\ln\left(x_2\right)^2 + bx_3\right)^5$$

$$y_6 = x_1 \mathrm{ch}\left(a\ln\left(x_2\right)^2 + bx_3\right)^2 + x_1 \sin\left(a\ln\left(x_2\right)^2 + bx_3\right) + \mathrm{sh}\left(a\ln\left(x_2\right)^2 + bx_3\right)^5$$

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Result:
Dependent variables:$s_9, s_{12}, y_6, s_7, s_2, s_6, s_{16}, s_8, s_{10}, y_5, s_1, y_3, s_4, s_{11}, y_2, s_{14}, s_{15}, s_{13}, y_1, y_4, s_5, s_3$
Independent variables:x_2, a, x_1, x_3, b
Additional variables:

$$s_1 = \ln\left(x_2\right)$$

$$s_2 = \mathrm{inv}\left(x_2\right)$$

$$s_3 = \cos\left(as_1^2 + bx_3\right)$$

$$s_4 = \sin\left(as_1^2 + bx_3\right)$$

$$s_5 = \sin\left(s_3\right)$$

$$s_6 = \cos\left(s_3\right)$$

$$s_7 = \mathrm{Dv}\left(g^2, 1, (-1)\,; x_1\right)$$

$$s_8 = \mathrm{DV}\left(g^2, 1, (-1)\,; x_1\right)$$

$$s_9 = \cos\left(s_4\right)$$

$$s_{10} = \sin\left(s_4\right)$$

$$s_{11} = \mathrm{EK}\left(s_{10}, x_1\right)$$

$$s_{12} = \mathrm{EKs}\left(s_{10}, x_1\right)$$

$$s_{13} = \mathrm{EKc}\left(s_{10}, x_1\right)$$

$$s_{14} = \mathrm{EKi}\left(s_{10}, x_1\right)$$

$$s_{15} = \mathrm{ch}\left(as_1^2 + bx_3\right)$$

$$s_{16} = \mathrm{sh}\left(as_1^2 + bx_3\right)$$

Functions:

$$y_1 = x_3^3 b s_1^4 + s_4^5 + x_3^3 s_5 + s_7$$

$$y_2 = b s_1^4 + s_4^5 + s_5 + s_7$$

$$y_3 = s_3^4 + x_2^2 s_9 + s_{11}^2$$

$$y_4 = s_3^4 + x_2^2 s_9 + s_{11}$$

$$y_5 = s_{16}^5 + x_1^2 s_{15}^2 + x_1^2 s_4$$

$$y_6 = s_{16}^5 + x_1 s_{15}^2 + x_1 s_4$$

Equations:

$$\frac{\partial s_1}{\partial x_2} = s_2$$

$$\frac{\partial s_1}{\partial a} = 0$$

$$\frac{\partial s_1}{\partial x_1} = 0$$

$$\frac{\partial s_1}{\partial x_3} = 0$$

$$\frac{\partial s_1}{\partial b} = 0$$

$$\frac{\partial s_2}{\partial x_2} = (-1)\, s_2^2$$

$$\frac{\partial s_2}{\partial a} = 0$$

$$\frac{\partial s_2}{\partial x_1} = 0$$

$$\frac{\partial s_2}{\partial x_3} = 0$$

$$\frac{\partial s_2}{\partial b} = 0$$

$$\frac{\partial s_3}{\partial x_2} = (-2)\, s_4 s_1 s_2 a$$

$$\frac{\partial s_3}{\partial a} = (-1)\, s_4 s_1^2$$

$$\frac{\partial s_3}{\partial x_1} = 0$$

$$\frac{\partial s_3}{\partial x_3} = (-1)\, s_4 b$$

$$\frac{\partial s_3}{\partial b} = (-1)\, s_4 x_3$$

$$\frac{\partial s_4}{\partial x_2} = 2 s_3 s_1 s_2 a$$

$$\frac{\partial s_4}{\partial a} = s_3 s_1^2$$

$$\frac{\partial s_4}{\partial x_1} = 0$$

$$\frac{\partial s_4}{\partial x_3} = s_3 b$$

$$\frac{\partial s_4}{\partial b} = s_3 x_3$$

$$\frac{\partial s_5}{\partial x_2} = (-2)\, s_6 s_4 s_1 s_2 a$$

$$\frac{\partial s_5}{\partial a} = (-1)\, s_6 s_4 s_1^2$$

$$\frac{\partial s_5}{\partial x_1} = 0$$

$$\frac{\partial s_5}{\partial x_3} = (-1)\, s_6 s_4 b$$

$$\frac{\partial s_5}{\partial b} = (-1)\, s_6 s_4 x_3$$

$$\frac{\partial s_6}{\partial x_2} = 2 s_5 s_4 s_1 s_2 a$$

$$\frac{\partial s_6}{\partial a} = s_5 s_4 s_1^2$$

$$\frac{\partial s_6}{\partial x_1} = 0$$

$$\frac{\partial s_6}{\partial x_3} = s_5 s_4 b$$

$$\frac{\partial s_6}{\partial b} = s_5 s_4 x_3$$

$$\frac{\partial s_7}{\partial x_2} = 0$$

$$\frac{\partial s_7}{\partial a} = 0$$

$$\frac{\partial s_7}{\partial x_1} = s_8$$

$$\frac{\partial s_7}{\partial x_3} = 0$$

$$\frac{\partial s_7}{\partial b} = 0$$

$$\frac{\partial s_8}{\partial x_2} = 0$$

$$\frac{\partial s_8}{\partial a} = 0$$

$$\frac{\partial s_8}{\partial x_1} = (-1)\, s_7 p_1 x_1^2 + (-1)\, s_7 p_2 x_1 + (-1)\, s_7 p_3$$

$$\frac{\partial s_8}{\partial x_3} = 0$$

$$\frac{\partial s_8}{\partial b} = 0$$

$$\frac{\partial s_9}{\partial x_2} = (-2)\, s_{10} s_3 s_1 s_2 a$$

$$\frac{\partial s_9}{\partial a} = (-1)\, s_{10} s_3 s_1^2$$

$$\frac{\partial s_9}{\partial x_1} = 0$$

$$\frac{\partial s_9}{\partial x_3} = (-1)\, s_{10} s_3 b$$

$$\frac{\partial s_9}{\partial b} = (-1)\, s_{10} s_3 x_3$$

$$\frac{\partial s_{10}}{\partial x_2} = 2 s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{10}}{\partial a} = s_9 s_3 s_1^2$$

$$\frac{\partial s_{10}}{\partial x_1} = 0$$

$$\frac{\partial s_{10}}{\partial x_3} = s_9 s_3 b$$

$$\frac{\partial s_{10}}{\partial b} = s_9 s_3 x_3$$

$$\frac{\partial s_{11}}{\partial x_2} = 2 s_{14} s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{11}}{\partial a} = s_{14} s_9 s_3 s_1^2$$

$$\frac{\partial s_{11}}{\partial x_1} = s_{12} s_{14}$$

$$\frac{\partial s_{11}}{\partial x_3} = s_{14} s_9 s_3 b$$

$$\frac{\partial s_{11}}{\partial b} = s_{14} s_9 s_3 x_3$$

$$\frac{\partial s_{12}}{\partial x_2} = 2 s_{13} s_{14} s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{12}}{\partial a} = s_{13} s_{14} s_9 s_3 s_1^2$$

$$\frac{\partial s_{12}}{\partial x_1} = s_{12} s_{13} s_{14}$$

$$\frac{\partial s_{12}}{\partial x_3} = s_{13} s_{14} s_9 s_3 b$$

$$\frac{\partial s_{12}}{\partial b} = s_{13} s_{14} s_9 s_3 x_3$$

$$\frac{\partial s_{13}}{\partial x_2} = (-2)\, s_{12} s_{14} s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{13}}{\partial a} = (-1)\, s_{12} s_{14} s_9 s_3 s_1^2$$

$$\frac{\partial s_{13}}{\partial x_1} = (-1)\, s_{12}^2 s_{14}$$

$$\frac{\partial s_{13}}{\partial x_3} = (-1)\, s_{12} s_{14} s_9 s_3 b$$

$$\frac{\partial s_{13}}{\partial b} = (-1)\, s_{12} s_{14} s_9 s_3 x_3$$

$$\frac{\partial s_{14}}{\partial x_2} = (-2)\, s_{10} s_{14}^3 s_{12} s_9 s_3 s_1 s_2 a$$

$$\frac{\partial s_{14}}{\partial a} = (-1)\, s_{10} s_{14}^3 s_{12} s_9 s_3 s_1^2$$

$$\frac{\partial s_{14}}{\partial x_1} = (-1)\, s_{10} s_{12}^2 s_{14}^3 + s_{13} s_{14}^2$$

$$\frac{\partial s_{14}}{\partial x_3} = (-1)\, s_{10} s_{14}^3 s_{12} s_9 s_3 b$$

$$\frac{\partial s_{14}}{\partial b} = (-1)\, s_{10} s_{14}^3 s_{12} s_9 s_3 x_3$$

$$\frac{\partial s_{15}}{\partial x_2} = 2 s_{16} s_1 s_2 a$$

$$\frac{\partial s_{15}}{\partial a} = s_{16} s_1^2$$

$$\frac{\partial s_{15}}{\partial x_1} = 0$$

$$\frac{\partial s_{15}}{\partial x_3} = s_{16} b$$

$$\frac{\partial s_{15}}{\partial b} = s_{16} x_3$$

$$\frac{\partial s_{16}}{\partial x_2} = 2 s_{15} s_1 s_2 a$$

$$\frac{\partial s_{16}}{\partial a} = s_{15} s_1^2$$

$$\frac{\partial s_{16}}{\partial x_1} = 0$$

$$\frac{\partial s_{16}}{\partial x_3} = s_{15} b$$

$$\frac{\partial s_{16}}{\partial b} = s_{15} x_3$$